

Theoretical Quantum Optics: Problem Sheet 6 (for 16.06.2025)

16. Decay of a macroscopic superposition state (1P)

Consider a harmonic oscillator that is described by the master equation

$$\dot{\rho} = \frac{\gamma}{2}(N_{\text{th}} + 1)(2a\rho a^\dagger - a^\dagger a \rho - \rho a^\dagger a) + \frac{\gamma}{2}N_{\text{th}}(2a^\dagger \rho a - aa^\dagger \rho - \rho aa^\dagger). \quad (1)$$

Assume that the harmonic oscillator is prepared in the pure state $\rho(t=0) = |\psi_0\rangle\langle\psi_0|$, where

$$|\psi_0\rangle = \frac{1}{\sqrt{2}}(|n\rangle + |m\rangle) \quad (2)$$

is a superposition of two number states. Calculate the initial decay of the coherence $c_{n,m}(t) = \langle n|\rho(t)|m\rangle$, i.e., show that

$$\dot{c}_{n,m}(t) = -\frac{\Gamma_{n,m}}{2}c_{n,m}(t), \quad \text{for } t \rightarrow 0. \quad (3)$$

Derive the rate $\Gamma_{n,m}$.

17. Decay of a macroscopic superposition state: numerics (2P)

Write a Python program to simulate the master equation of a damped harmonic oscillator given in Eq. (1).

(a) Calculate the exact decay of $c_{n,m}(t)$ and compare it with the approximate expression

$$c_{n,m}(t) \simeq \frac{1}{2}e^{-\Gamma_{n,m}t/2}.$$

Choose $N_{\text{th}} \lesssim 5$ to keep the dimension of the Hilbertspace reasonably small.

(b) Evaluate $\rho(t)$ for times $\gamma t = [0.05, 0.5, 5]$ and $N_{\text{th}} = 3$ and visualize the decay of the initial superposition state into a thermal equilibrium state. Use, for example, the ‘pcolor’ function from matplotlib to show all the elements of the density operator in matrix form.

18. Quantum Regression Theorem (2P)

Consider a damped cavity mode, which is described by the master equation

$$\dot{\rho} = -i[H, \rho] + \frac{\gamma}{2}(N_{\text{th}} + 1)(2a\rho a^\dagger - a^\dagger a \rho - \rho a^\dagger a) + \frac{\gamma}{2}N_{\text{th}}(2a^\dagger \rho a - aa^\dagger \rho - \rho aa^\dagger) \quad (4)$$

with Hamiltonian

$$H = i\hbar\frac{g}{2}(a^2 - a^{\dagger 2}) + i\Omega(a - a^\dagger), \quad (5)$$

It is assumed that $g < \gamma/2$.

- (a) Derive the equations of motion for the expectation values of $\langle a(t) \rangle$ and $\langle a^\dagger(t) \rangle$. Based on this result, use the quantum regression theorem to derive a closed set of equations of motion for the correlation functions $\langle a^\dagger(\tau)a(0) \rangle$ and $\langle a(\tau)a(0) \rangle$.
- (b) Set $g = \Omega = 0$ and consider the correlation function $G^{(2)}(\tau) = \langle a^\dagger(0)a^\dagger(\tau)a(\tau)a(0) \rangle$. What is the value of $G^{(2)}(0) = 2N_{\text{th}}^2$ for a thermal equilibrium state? Use the quantum regression theorem to derive an expression for $G^{(2)}(\tau)$ for arbitrary τ .

19. Mollow triplet (1P)

By defining the vector of Pauli operators $\vec{S} = (\sigma_-, \sigma_+, \sigma_z)$, the optical Bloch equations for a driven two-level atom can be written as

$$\partial_t \langle \vec{S} \rangle = \mathbf{M} \langle \vec{S} \rangle - \vec{R}, \quad (6)$$

with a matrix $\mathbf{M}$ and a vector $\vec{R}$ derived in the lecture. With this definition the spectrum of incoherently scattered photons is given by

$$S_{\text{inc}}(\omega) = 2\Gamma \text{Re}(0, 1, 0) \left[\frac{1}{i(\omega - \omega_L)\mathbb{1} - \mathbf{M}} \right] \vec{C}(\tau = 0), \quad (7)$$

where $\vec{C}(0) = \langle \vec{S}\sigma_- \rangle_0 - \langle \vec{S} \rangle_0 \langle \sigma_- \rangle_0$ and $\langle \cdot \rangle_0$ denotes the steady-state expectation value.

- (a) Write a Python program to evaluate and plot $S_{\text{inc}}(\omega)$ for arbitrary system parameters.
- (b) Plot $S_{\text{inc}}(\omega)$ for a resonantly driven atom, $\Delta = 0$, and for increasing values of $\Omega/\Gamma = 0.1, 1, 2, 5, 10$.
- (c) Plot and discuss $S_{\text{inc}}(\omega)$ for $\Omega/\Gamma \approx 5$ and different values of $\Delta/\Gamma = 0, 5, 10$. How do the positions and relative weights of the scattering peaks depend on the detuning?

Hint: For this exercise you can also use the code from the ‘Quantum Optics with Python’ notes as a starting point.

$$16) \dot{\phi}(t) = \frac{\gamma}{2} (N_{th} + 1) (2a^\dagger p a^\dagger - a^\dagger a p - p a^\dagger a) + \frac{\gamma}{2} N_{th} (2a^\dagger p a - a a^\dagger p - p a a^\dagger)$$

$$\langle n | \dot{\phi} | m \rangle = \frac{\gamma}{2} (N_{th} + 1) \left(2\sqrt{n+1}\sqrt{m+1} c_{n+1,m+1} - \sqrt{n}\sqrt{n} c_{n,m} - \sqrt{m}\sqrt{m} c_{n,m} \right) + \frac{\gamma}{2} N_{th} \left(2\sqrt{n}\sqrt{m} c_{n,m} - (n+1) c_{n,m} - c_{n,m} (m+1)^2 \right)$$

$\uparrow$
 ~ 0 for $|1\rangle = \frac{1}{\sqrt{2}}(|n\rangle + |m\rangle)$ and $t \rightarrow 0$

$$= -\frac{\gamma}{2} (N_{th} + 1) (n+m) c_{n,m} - \frac{\gamma}{2} N_{th} (n+1+m+1)$$

$$= \left[-\frac{\gamma}{2} (N_{th} + 1) (n+m) - \frac{\gamma}{2} (n+m) N_{th} \right] c_{n,m}$$

$$-\frac{\Gamma_{n,m}}{2} \rightarrow \Gamma_{n,m} = \gamma \left[(2N_{th} + 1)(n+m) + 2N_{th} \right]$$

18.) $\langle a(t) \rangle = \text{tr}(\rho(t) a(t)) \quad (t=1)$

$$\langle \dot{a}(t) \rangle = i \langle [H, a] \rangle + \frac{\gamma(N_{th}+1)}{2} \langle [a^\dagger, a] a + a^\dagger [a, a^\dagger] \rangle + \frac{\gamma N_{th}}{2} \langle [a, a^\dagger] a^\dagger + a [a, a^\dagger] \rangle$$

$$= [i \frac{\gamma}{2} (a^2 + a^{\dagger 2}), a] + i\Omega [a - a^\dagger, a]$$

$$= i \frac{\gamma}{2} ([a^2, a] + [a^{\dagger 2}, a]) + i\Omega (-[a^\dagger, a])$$

$$= i \frac{\gamma}{2} (a^\dagger [a^\dagger, a] + [a^\dagger, a] a) + i\Omega$$

$$= i \frac{\gamma}{2} (-2a^\dagger) + i\Omega = i(\Omega - \gamma a^\dagger)$$

$$= -\Omega + \gamma \langle a^\dagger \rangle - \frac{\gamma(N_{th}+1)}{2} \langle a \rangle + \frac{\gamma N_{th}}{2} \langle a \rangle$$

$$= -\Omega + \gamma \langle a^\dagger \rangle - \frac{\gamma}{2} \langle a \rangle$$

$$\Rightarrow \langle \dot{O} \rangle(t) = \frac{i}{\hbar} \langle [H, O] \rangle + \sum_k \frac{\Gamma_k}{2} \langle [c_k^\dagger, O] c_k + c_k^\dagger [O, c_k] \rangle$$

$$C_0 = a \quad C_1 = a^\dagger$$

$$\Gamma_0 = \gamma(N_{th}+1) \quad \Gamma_1 = \gamma N_{th}$$

$$\langle a^\dagger(t) \rangle = i \langle [H, a^\dagger] \rangle + \frac{\gamma(N_{th}+1)}{2} \langle a^\dagger [a^\dagger, a] \rangle + \frac{\gamma N_{th}}{2} \langle [a, a^\dagger] a^\dagger \rangle = -\gamma \langle a \rangle - \Omega - \frac{\gamma}{2} \langle a^\dagger \rangle$$

$$\hookrightarrow \begin{aligned} &= [i \frac{\gamma}{2} (a^2 + a^{\dagger 2}), a^\dagger] + i\Omega [a - a^\dagger, a^\dagger] \\ &= i \frac{\gamma}{2} ([a^2, a^\dagger] + [a^{\dagger 2}, a^\dagger]) + i\Omega ([a, a^\dagger]) \\ &= i \frac{\gamma}{2} (a [a, a^\dagger] + [a, a^\dagger] a) + i\Omega \\ &= i \frac{\gamma}{2} (2a) + i\Omega = i(\Omega + \gamma a) \end{aligned}$$

$$\Rightarrow \frac{d}{dt} \begin{pmatrix} \langle a \rangle \\ \langle a^\dagger \rangle \end{pmatrix} = \begin{pmatrix} -\gamma/2 & \gamma \\ -\gamma & -\gamma/2 \end{pmatrix} \begin{pmatrix} \langle a \rangle \\ \langle a^\dagger \rangle \end{pmatrix} - \Omega \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \frac{d}{d\tau} \langle a^\dagger(\tau) a(0) \rangle = -\gamma \langle a(\tau) a(0) \rangle - \frac{\gamma}{2} \langle a^\dagger(\tau) a(0) \rangle - \Omega \langle a(0) \rangle$$

from regression theorem!

$$\frac{d}{d\tau} \langle a(\tau) a(0) \rangle = -\frac{\gamma}{2} \langle a(\tau) a(0) \rangle + \gamma \langle a^\dagger(\tau) a(0) \rangle - \Omega \langle a(0) \rangle$$

$$\Rightarrow \frac{d}{d\tau} \begin{pmatrix} \langle a(\tau) a(0) \rangle \\ \langle a^\dagger(\tau) a(0) \rangle \end{pmatrix} = \begin{pmatrix} -\frac{\gamma}{2} & \gamma \\ -\gamma & -\frac{\gamma}{2} \end{pmatrix} \begin{pmatrix} \langle a(\tau) a(0) \rangle \\ \langle a^\dagger(\tau) a(0) \rangle \end{pmatrix} - \Omega \begin{pmatrix} 1 \\ 1 \end{pmatrix} \langle a(0) \rangle$$

b) $G^{(2)}(\tau) = \langle a^\dagger(0) a^\dagger(\tau) a(\tau) a(0) \rangle = \langle a^\dagger(0) a^\dagger(\tau) a(0) a(\tau) \rangle$

$H=0$ for $\gamma=\Omega=0$

since $a(\tau) = a(0) e^{-\gamma\tau/2}$
from above

$$= \langle a^\dagger(0) a(0) a^\dagger(\tau) a(\tau) \rangle = e^{-\frac{\gamma}{2}\tau} \langle a^\dagger(0) a(\tau) \rangle$$

$$[a^\dagger(\tau), a(0)] = [a^\dagger(0) e^{-\frac{\gamma}{2}\tau}, a(0)] = -e^{-\frac{\gamma}{2}\tau}$$

$$= \langle n(0) n(\tau) \rangle = e^{-\frac{\gamma}{2}\tau} \langle a^\dagger(0) a(\tau) \rangle$$

from above and $\langle a^\dagger(0) a(0) \rangle = N_{th}$ initial cond.
we get fixed expression: $N_{th} e^{-\gamma\tau/2}$

$$\langle n(0) n(\tau) \rangle: \frac{d}{d\tau} \langle n \rangle = \frac{\gamma(N_{th}+1)}{2} \langle [a^\dagger, n] a + a^\dagger [n, a] \rangle + \frac{\gamma N_{th}}{2} \langle [a, n] a^\dagger + a [n, a^\dagger] \rangle$$

$$= \frac{\gamma(N_{th}+1)}{2} (-\langle a^\dagger a \rangle - \langle a^\dagger a \rangle) + \frac{\gamma N_{th}}{2} (\langle a a^\dagger \rangle + \langle a a^\dagger \rangle)$$

$$= -\gamma(N_{th}+1) \langle n \rangle + \gamma N_{th} \langle n \rangle$$

$$= -\gamma N_{th} \langle n \rangle \rightarrow \text{regression: } \frac{d}{d\tau} \langle n(0) n(\tau) \rangle = -\gamma \langle n(0) n(\tau) \rangle + \gamma N_{th} \langle n(0) \rangle$$

$$\Rightarrow \text{with } \langle n^2(0) \rangle = \underbrace{\text{Var}(n)}_{N_{th}(N_{th}+1)} + \underbrace{\langle n(0) \rangle^2}_{N_{th}^2}$$

$$= 2N_{th}^2 + N_{th}$$

$$\Rightarrow \langle n(0) n(\tau) \rangle = N_{th}^2 + (N_{th}^2 + N_{th}) e^{-\gamma\tau}$$

$$\Rightarrow G^2(\tau) = N_{th}^2 + (N_{th}^2 + N_{th})e^{-\gamma\tau} - e^{-\gamma\tau}N_{th}$$

$$= N_{th}^2(1 + e^{-\gamma\tau})$$

$$\Rightarrow G^2(0) = N_{th}^2(1 + 1) = 2N_{th}^2 \quad \square$$